

BIO504T SUBJECTIVES

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BIOCHEMISTRY I

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SUBJECTIVES

How fireflies produce light?

The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase.

Difference between Bees wax and spermaceti:

Bees-wax is secreted by the honeybees that use it to form the combs. It is a mixture of waxes. chief constituent is myricyl (30C), palmitate (16C).

Spermaceti is a wax that is primarily found in the head cavities of sperm whales. It consists of fatty esters, mainly cetyl palmitate and cetyl myristate. Spermaceti has been used in cosmetics, pharmacy, and candles, but its use has been restricted due to international regulations on whale captures. Synthetic cetyl palmitate is now used as a replacement.

Properties of lipid compounds:

Lipid compounds, specifically fatty acids, exhibit poor water solubility due to nonpolar hydrocarbon chains. Solubility decreases with larger acyl chains and fewer double bonds. Formation of esters, salts (soaps), and detergents are notable properties, highlighting their diverse chemical interactions and applications.

Effect of non-competitive inhibitors on V_{max} :

Noncompetitive inhibitors decrease V_{max} by preventing catalysis independent of substrate binding. Their impact is irreversible and cannot be overcome by increasing substrate concentration. The inhibitor reduces the active enzyme concentration, ultimately lowering the maximum reaction rate (V_{max}).

Define Selenocysteine, formation and function.

Selenocysteine is an L- α -amino acid present in select proteins like peroxidases and reductases, crucial for electron transfer reactions. Its synthesis involves modifying a serine bound to unique RNA, where the hydroxyl group is replaced by a selenium atom. This substitution occurs during translation, earning selenocysteine the label of the "21st amino acid." Unlike others, it lacks a standard three-letter codon, making its incorporation distinctive and underlining its role in catalyzing essential reactions during protein synthesis.

What is 2,3 bisphosphoglycerate and explain.

2,3-bisphosphoglycerate (BPG) is a byproduct of glycolysis and plays a pivotal role in hemoglobin function as a heterotropic allosteric modulator. In conditions of low oxygen delivery to tissues, BPG levels rise in red blood cells. Negatively charged, it stabilizes the deoxygenated form of hemoglobin by binding to a cavity between beta subunits, reducing the protein's affinity for oxygen. This adjustment, particularly at high altitudes, facilitates increased oxygen unloading in peripheral tissues, highlighting BPG's crucial role in physiological adaptation and oxygen delivery regulation.

Properties of nitrogenous Bases :

1. **Aromatic Nature:** Nitrogenous bases exhibit an aromatic character with alternate double bonds, contributing to their molecular structure.
2. **Heterocyclic Structure:** These bases are heterocyclic, containing atoms beyond carbon, such as nitrogen, in their ring structures.
3. **Weak Bases:** Purines or pyrimidines possessing an $-NH_2$ group act as weak bases.
4. **Key Functional Groups:** Prominent functional groups include ring nitrogens, carbonyl groups, and exocyclic amino groups.
5. **Hydrophobicity:** Purine and pyrimidine bases demonstrate hydrophobicity, making them relatively insoluble in near-neutral cell pH water.
6. **Stacking Interaction:** Hydrophobic stacking interactions, resembling a stack of coins, represent a significant mode of interaction between nucleic acid bases.

Write inorganic compound serve as cofactor:

The inorganic cofactor involved in coenzymes include:

1= Zinc (Zn), 2= Iron (Fe), 3= Magnesium (Mg), 4= Copper (Cu), 5= Manganese (Mn).

Three names of polyunsaturated fatty acid:

Polyunsaturated fatty acids contain more than one double bond, These include:

- 1) Linoleic acid (18: 2 Δ 9, 12)
- 2) α -Linolenic acid (18 : 3 Δ 9, 12, 15)
- 3) Arachidonic acid series (20 : 4 Δ 5, 8, 11, 14)

Properties of proteins:

Proteins exhibit diverse properties crucial for biological functions, including enzymatic catalysis, structural support, and cellular signaling. Their properties depend on the amino acid sequence, folding patterns, and interactions. Proteins are sensitive to environmental factors, such as pH and temperature, and their functionality relies on proper folding and structural integrity.

Primary structure of protein:

The primary structure of a protein refers to its linear sequence of amino acids, determined by the specific order of peptide bonds. It serves as the foundation for higher-order structural configurations in proteins.

What are the different classifications of lipids.

Lipids are classified into three groups:

1. Simple Lipids:

- ❖ **Fats (and Oils):** Esters of fatty acids with glycerol. Oils refer to fats in the liquid state.
- ❖ **Waxes:** Esters of fatty acids with higher molecular weight monohydric alcohols.

2. Complex Lipids:

- ❖ **Phospholipids:** Contain fatty acids, alcohol, and a phosphoric acid residue.
- ❖ **Glycolipids (Glycosphingolipids):** Include fatty acid, sphingosine, and a carbohydrate.
- ❖ **Other Complex Lipids:** Encompass sulfolipids, aminolipids, and lipoproteins.

3. Precursor and Derived Lipids:

- ❖ Include fatty acids, glycerol, steroids, other alcohols, fatty aldehydes, ketone bodies, and lipid-soluble vitamins.

What are the Inhibitors and Classes of Enzyme Inhibitors:

Enzyme inhibitors are molecular agents that impede enzymatic reactions, playing a crucial role in pharmaceutical agents. Classified based on their site of action, chemical modification, or kinetic parameters, inhibitors fall into two main classes: reversible and irreversible. Irreversible inhibitors form covalent bonds with enzymes, while reversible inhibitors, the most common being competitive and noncompetitive, utilize noncovalent bonds. Competitive inhibitors resemble the substrate, competing for the enzyme's active site, while noncompetitive inhibitors bind elsewhere, either to free enzyme or the enzyme-substrate complex. Aspirin exemplifies an inhibitor affecting prostaglandin synthesis.

What is meant by protein folding?

Protein folding refers to the process by which a linear chain of amino acids assumes its three-dimensional, functional structure. It involves intricate spatial arrangements, driven by interactions such as hydrogen bonding, hydrophobic forces, and disulfide bonds, ensuring proper protein functionality.

What is the Condensation reaction with examples:

A condensation reaction is a chemical process where two molecules combine, eliminating a small molecule like water or alcohol. This results in the formation of a larger molecule. An example is the synthesis of an ester from an alcohol and carboxylic acid, releasing water in the process. Other examples: N-glycosidic bond, O-glycosidic bond.

Sphingomyelin and sub classes of sphingolipids:

Sphingomyelin, a significant sphingophospholipid in humans, belongs to the subclass of sphingolipids known as sphingomyelins. It features phosphorylcholine esterified to the alcohol group at carbon 1 of sphingosine. Predominantly present in animal cell plasma membranes and nerve tissues, especially myelin, sphingomyelins contribute to myelin sheath structure. Sphingolipids encompass two subclasses: Sphingomyelins, containing phosphocholine or phosphoethanolamines, and Sphingoglycolipids, complex molecules with both carbohydrate and lipid components, emphasizing their diverse roles in cellular membranes and neurological functions.

Five properties of waxes:

1= Waxes are insoluble in water, but soluble in fat, 2= Solvents and are negative for acrolein test, 3= Very resistant to rancidity, 4= Waxes are not easily hydrolyzed as the fats, 5= waxes are indigestible by lipases (enzymes responsible for fat Digestion in body) Thus they are of no nutritional value.

What is the Cis-isomerism:

Cis-isomerism refers to the arrangement of atoms or groups on the same side of a double bond in a molecule. It is important because it affects the physical and chemical properties of the molecule, such as its melting point, boiling point, and reactivity.

Rancidity and factors causing rancidity?

It is a physico-chemical change in the natural properties of the fat leading to the development of unpleasant odor or taste or abnormal color. It occurs particularly

- ❖ on aging
- ❖ after exposure to atmospheric oxygen,
- ❖ light,
- ❖ moisture,
- ❖ bacterial or fungal contamination
- ❖ and/or heat.

Rancidity is due to

- I. Oxidation
- II. Hydrolysis

Difference between oxidative and hydrolytic rancidity.

Oxidative Rancidity

Oxidative rancidity occurs when fats undergo oxidation, resulting in the formation of aldehydes, ketones, and dicarboxylic acids that contribute to an objectionable taste and odor. This type of rancidity requires the presence of oxygen in the air and can be prevented by adding antioxidants like vitamin E to foods.

Hydrolytic Rancidity

Hydrolytic rancidity, on the other hand, is caused by the slow hydrolysis of fats. In the case of fats like butter, this leads to the release of short-chain fatty acids that are volatile and have a rancid taste and odor. Unlike oxidative rancidity, hydrolytic rancidity does not require the presence of oxygen.

Function of m RNA:

Messenger RNA (mRNA): mRNA carries the genetic information from DNA to the ribosomes, where it serves as a template for protein synthesis.

What is the composition of bile?

The composition of gallbladder bile is 97% water, 0.7% bile salts, 0.2% bilirubin, 0.51% fats (cholesterol, fatty acids, and lecithin), and 200 meq/l inorganic salts.

How activation energy related to enzyme:

Enzymes lower the activation energy required for a chemical reaction to occur. Activation energy is the energy needed to start a reaction and break the existing bonds in the reactant molecules. Enzymes facilitate this process by binding to the reactant molecules, creating an enzyme-substrate complex. This complex lowers the activation energy by stabilizing the transition state, allowing the reaction to proceed more quickly.

Palmitic Acid(16:0) Characteristics:

- 1= Palmitic acid, or hexadecanoic acid, is the most common saturated Fatty Acid found in animals, plants and microorganisms.
- 2= Palmitic acid mainly occurs as its ester in triglycerides (fats), especially in palm oil.
- 3= It is also found in high amounts in • Butter, • Cheese, • milk and • meat,
- 4= Excess carbohydrates in the body are converted to palmitic acid.
- 5= Palmitic acid is the first fatty acid produced during fatty acid synthesis and the precursor to longer fatty acids.

Primary Structure Of DNA:

In DNA double helix, the two strands of DNA are held together by hydrogen bonds. The primary structure of DNA refers to its covalent structure and nucleotide sequence. It is a linear polymer of deoxyribonucleotides (dNTPs) linked by 5'-3' phosphodiester bonds, forming an unbranched chain. The backbone comprises sugar-phosphate residues, creating a long, linear structure with polarity. Both the 5'-end and 3'-end are free, featuring a phosphate group at the 5'-end and a free hydroxyl group at the 3'-end. The variable part, formed by purines and pyrimidines projecting laterally, is crucial for expressing genetic information.

DNA Renaturation /Intact Duplex:

Under appropriate conditions (temp. & salt concentration), separated strands of DNA will renature or reassociate and form the double helix by the process called renaturation (or reannealing). This reannealing process is also referred to as hybridization. When the temperature or pH is returned to the range in which most organisms live, the unwound segments of the two strands spontaneously rewind, or anneal, to yield the intact duplex.

Zero Order Reaction:

In chemistry, a zero-order reaction is a type of reaction where the rate of the reaction is constant and independent of the concentration of the reactant(s). In other words, the reaction proceeds at a fixed rate, regardless of how much reactant is present. The rate equation for a zero-order reaction can be represented as "Rate = k", where "k" is the rate constant. This means that the concentration of the reactant(s) does not affect the rate of the reaction.

Factors Affecting Enzymatic Activity:

Different enzymes show different responses to changes in; 1= substrate concentration, 2= temperature, and, 3= pH. Enzymic responses to these factors give us valuable clues as to how enzymes function in living cells (that is, in vivo).

Define Four levels of proteins:

Proteins exhibit four levels of structural organization: Primary structure involves the linear sequence of amino acids in a polypeptide chain. Secondary structure results from interactions such as alpha-helices and beta-sheets, folding due to hydrogen bonding. Tertiary structure represents the overall three-dimensional arrangement of a single polypeptide chain. Quaternary structure involves the interaction of multiple polypeptide chains in a protein complex. These hierarchical levels collectively determine a protein's function and are crucial for its stability and biological activity.

Define the quaternary structure of protein:

Quaternary structure involves the interaction of multiple polypeptide chains in a protein complex.

Numbering of Carbon and Nitrogen Atoms:

In nitrogenous bases, carbons in the pyrimidine ring are numbered 1 to 6, and in purines, they are numbered 1 to 9. In pentoses of nucleotides, carbon numbers are designated 1' to 5', with primes distinguishing them from the numbered atoms of the nitrogenous base.

Features or characteristics of VLDL:

1= VLDL is a type of lipoprotein that transports triglycerides from the liver to peripheral tissues. 2= It is composed of triglycerides, cholesterol, and proteins. 3= VLDL is a precursor to LDL (Low-Density Lipoprotein), also known as "bad cholesterol." 4= High levels of VLDL in the blood are associated with an increased risk of cardiovascular disease. 5= VLDL levels can be influenced by factors such as diet, genetics, and certain medical conditions. 6= VLDL is synthesized in the liver and released into the bloodstream. 7= It provides energy to cells by delivering triglycerides to tissues for storage or immediate use. 8= VLDL particles are larger and

less dense than LDL particles. 9= VLDL levels can be measured through a blood test called a lipid profile. 10= Managing VLDL levels through lifestyle changes, such as a healthy diet and regular exercise, can help maintain cardiovascular health.

Explain types of RNA:

There are several types of RNA that play different roles in cellular processes: 1. Messenger RNA (mRNA): mRNA carries the genetic information from DNA to the ribosomes, where it serves as a template for protein synthesis. 2. Transfer RNA (tRNA): tRNA molecules bring amino acids to the ribosomes during protein synthesis. Each tRNA molecule is specific for a particular amino acid. 3. Ribosomal RNA (rRNA): rRNA is a major component of ribosomes, which are the cellular machinery responsible for protein synthesis. It helps in the assembly of amino acids into polypeptide chains. 4. Small nuclear RNA (snRNA): snRNA is involved in RNA splicing, a process that removes noncoding regions (introns) from pre-mRNA to produce mature mRNA.

How temperature affect the enzyme activity:

The reaction velocity increases with temperature until a peak velocity is reached • This increase is the result of the increased number of molecules having sufficient energy to pass over the energy barrier and form the products. Further elevation of the temperature results in a decrease in reaction velocity as a result of temperature-induced denaturation of the enzyme.

What are the Lipoxins:

Lipoxins, an acronym for lipoxygenase interaction products, are bioactive autacoid metabolites of arachidonic acid made by various cell types.

Nomenclature of Prostaglandins (PGs):

Prostaglandins are fatty acids containing 20 carbon atoms, including an internal 5- carbon ring.

Nomenclature of Prostaglandins (PGs) involves the assignment of a capital letter (PGE), a numeral subscript (PGE1), and for the PGF family, a Greek letter subscript (PGF2 α). The capital letter refers to the ring substituents at positions X and Y. The numeral subscript that follows the capital letter(e.g. PGE1) refers to the PG series 1, 2, or 3, determined by the number of unsaturated bonds present in the linear portion of the hydrocarbon chain. It does not include double bonds in the internal ring. The double bonds between carbons 13 and 14 are trans; the others are cis. The Greek letter subscript, found only in the F series, refers to the position of the hydroxyl group at carbon 9. This hydroxyl group primarily exists in the alpha position, i.e. it lies below the plane of the ring.

What are the Properties of histidine:

Histidine is an essential amino acid with unique properties. It contains an imidazole side chain, making it a key player in enzymatic catalysis and metal binding. Its pKa near physiological pH enables buffering, crucial for maintaining stable cellular environments. Additionally, histidine plays a role in proton transfer reactions.

Define alpha, beta and gama carbon:

In organic chemistry, the alpha carbon refers to the carbon adjacent to a functional group in a molecule. The beta carbon is the second carbon, and the gamma carbon is the third carbon away from the functional group. These designations help describe the positions of atoms in relation to the functional group and are commonly used in nomenclature and discussions of chemical structures.

Adenine guanine difference:

Adenine and guanine are purine bases in nucleic acids. The primary difference lies in their structures. Adenine has a two-ring structure, while guanine possesses an additional carbonyl

group, forming a three-ring structure. This distinction influences their base-pairing specificities in DNA and RNA.

DNA structure, palmitic acid, hexokinase:

DNA Structure: DNA is a double-stranded helical molecule comprising nucleotide units with adenine, thymine, cytosine, and guanine bases. The strands run antiparallel, held together by complementary base pairing (A-T and G-C).

Palmitic Acid: Palmitic acid is a saturated fatty acid with a 16-carbon chain, abundant in animal fats and palm oil. Its structure includes a carboxyl group attached to a long hydrophobic hydrocarbon tail.

Hexokinase: Hexokinase is an enzyme catalyzing the phosphorylation of glucose to glucose-6-phosphate, initiating glycolysis.

Beta sheet and amino acid:

Beta sheets are secondary structures in proteins formed by hydrogen-bonded strands. Amino acids are the building blocks of proteins, and their sequence determines the protein's primary structure, including the arrangement of beta sheets.

Physical properties of amino acid:

Amino acids possess distinctive physical properties. They are water-soluble due to polar side chains, exhibit varying solubilities in organic solvents, and can form zwitterions. Their isoelectric point is where they are least soluble. Amino acids also have diverse melting points, reflecting their individual chemical structures and interactions.

Characteristics of furfural compound

Furfural is a heterocyclic aldehyde derived from agricultural residues. It possesses a sweet, almond-like odor, contributing to its applications in flavorings and perfumes. Furfural is a colorless liquid with a boiling point of 161.7°C. It is a versatile chemical intermediate used in the synthesis of resins, solvents, and pharmaceuticals.

Note on ligand:

A ligand, in molecular terms, is any molecule, including proteins, that binds reversibly to another molecule, typically a protein. The transient nature of these interactions is vital for rapid and reversible responses to environmental changes. Ligands can be small molecules that bind to proteins through weak interactions like ionic bonds, hydrogen bonds, Van der Waals forces, and hydrophobic effects. They exhibit specificity by complementing the protein's binding site in size, shape, charge, and hydrophobicity, performing roles such as substrate, inhibitor, activator, or neurotransmitter.

Difference between Thymine and uracil structure:

Thymine and uracil are nitrogenous bases in nucleic acids. The primary structural difference lies in thymine's methyl group at the 5th carbon, absent in uracil. Both are pyrimidines, but thymine is specific to DNA, while uracil replaces it in RNA.

Iodine number & its importance?

Iodine number is defined as the number of grams of iodine absorbed by 100 gm of fat.

Importance: The iodine number is important because it helps determine the degree of unsaturation in fats, which can indicate the quality of an oil and its susceptibility to oxidation and rancidification. It also helps identify potential adulteration in oils.

Standard structure of RNA:

The standard structure of RNA consists of a single-stranded, linear molecule composed of nucleotides. Each nucleotide comprises a phosphate group, ribose sugar, and one of four bases:

adenine, guanine, cytosine, or uracil. RNA is involved in diverse cellular functions, such as protein synthesis, catalysis, and gene regulation.

DNA interaction:

DNA interactions involve hydrophobic stacking where purine and pyrimidine bases, being hydrophobic, align with parallel planes. This stacking minimizes water contact and is crucial for stabilizing nucleic acid structures. Additionally, hydrogen bonds, specifically A-T (or U) and G-C, form the second essential interaction, defining the double-stranded DNA and RNA structures proposed by Watson and Crick.

Enzyme classification:

Enzymes are classified based on the type of reaction they catalyze. Six main categories include oxidoreductases, transferases, hydrolases, lyases, isomerases, and ligases. Each class has specific functions, playing essential roles in diverse biochemical processes.

Triacylglycerol role in plant:

Triacylglycerols (TAGs) in plants serve as crucial energy storage, stored as oils in seeds. During seed germination, lipases break down TAGs into fatty acids, releasing energy and building blocks for biosynthesis. This process supports plant growth, development, and metabolic processes by transporting fatty acids to areas where needed.

TAGs in detail:

TAGs (Triacylglycerols) are neutral fats, colorless, odorless, and tasteless, insoluble in water but soluble in organic solvents. They have a specific gravity less than 1.0, causing them to float in water. Emulsification of TAGs is vital for digestion and absorption, achieved by bile salts in the small intestine. These salts break apart fat globules, making them more soluble, facilitated by emulsifiers, ensuring effective utilization of dietary fats in the digestive system.

Triglycerides mixed names:

Mixed triglycerides have different types of fatty acids. Examples include

- Stearo-diolein
- Palmito-oleo-stearin
- Oleo-palmito-stearin
- Lauric-myristic-palmitic
- Oleo-diolein

Eicosanoid:

Eicosanoids are a large group of lipid messengers with potent effects on every tissue in the body. Eicosanoids are derived from metabolism of 20- carbon, polyunsaturated fatty acids (eicosanoic acids). Eicosanoids are derived from either Omega-3 (ω -3) or Omega-6 (ω -6) fatty acids. Arachidonic acid is the most common precursor of the eicosanoids.

What is term induced fit responsible binding of protein?

Induced fit is a concept in enzymology where the binding of a substrate to a protein causes a conformational change in the protein's active site. This change enhances binding affinity and optimizes the alignment of the substrate for catalysis, ensuring specificity and efficiency in enzymatic reactions.

Uses of lanolin?

Lanolin, derived from sheep sebaceous glands, contains cholesterol esters like cholesterol palmitate. It serves multiple purposes, including reducing water evaporation from sheep skin. Lanolin finds applications in industrial lubricants and cosmetics due to its moisturizing properties.

Nucleotides as regulatory compounds... Explain it...

Nucleotides are termed regulatory compounds as they play pivotal roles in metabolic regulation. They engage in ATP-dependent enzyme phosphorylation, allosteric regulation by ATP, AMP, and CTP, and modulation of enzyme activity through phosphorylation or dephosphorylation, either activating or inhibiting ATP-dependent enzymes based on the enzyme type.

Lineweaver burk plot explain it...

The Lineweaver-Burk plot is a graphical representation of the Michaelis-Menten equation, displaying a double reciprocal relationship between the initial reaction rate ($1/V_0$) and substrate concentration ($1/[S]$). By transforming the Michaelis-Menten equation, this plot provides a straight line, making it valuable for determining kinetic parameters like K_m and V_{max} accurately. It is especially useful for analyzing enzyme inhibition and discerning various enzymatic reaction mechanisms.

What is carbohydrates? four group of carbohydrates?

Carbohydrates are polyhydroxy aldehydes or ketones, or substances yielding such compounds upon hydrolysis. They often have the empirical formula $(CH_2O)_n$, with some containing nitrogen, phosphorus, or sulfur. Carbohydrates are categorized into four groups: Monosaccharides (e.g., glucose), Oligosaccharides (short chains of 3-10 monosaccharide units), Disaccharides (e.g., maltose, yielding two monosaccharide molecules upon hydrolysis), and Polysaccharides or Glycans (yielding more than 10 monosaccharide molecules upon hydrolysis, with some having linear chains like cellulose and others being branched like glycogen).

Function of cyclic AMP:

Cyclic AMP (cAMP), a cyclic nucleotide derived from ATP, serves as a second messenger within cells. Its functions include playing a crucial role in glycogen metabolism, promoting glycogenolysis and triglyceride metabolism, and enhancing lipolysis. cAMP also reduces cholesterol synthesis, activates protein kinases influencing enzyme activity, and regulates cell membrane permeability to water, Na^+ , K^+ , and Ca^{2+} . Additionally, it modulates insulin secretion, catecholamine biosynthesis, and melatonin synthesis, showcasing its diverse regulatory roles in cellular processes.

Large and small non coding regulatory RNA:

Large and small noncoding regulatory RNAs play vital roles in eukaryotic regulatory biology. Small noncoding RNAs, such as microRNAs (miRNAs) and silencing RNAs (siRNAs), inhibit gene expression by targeting mRNAs through distinct mechanisms. On the other hand, large noncoding RNAs (lncRNAs) contribute to regulation without encoding proteins. These discoveries, made in the past decade, highlight the significance of noncoding RNAs in controlling gene expression and influencing various cellular processes.

Define amylase

Amylase is an enzyme that catalyzes the hydrolysis of starch and glycogen into simpler sugars like maltose and glucose. Found in saliva and pancreatic secretions, amylase plays a crucial role in breaking down complex carbohydrates during digestion.

Which amino acids present in beta bends?

Glycine and Proline residues are commonly found in beta-bends. Glycine's small and flexible nature, along with Proline's propensity for assuming the cis configuration in peptide bonds involving the imino nitrogen, makes them suitable for forming the tight turns characteristic of beta-bends.

Name three hypercholesterolemic acids?

Lauric (12:0), myristic (14:0), and palmitic (16:0) acids are considered to be hypercholesterolemic.

Functional classification of proteins.

Functional classification of proteins categorizes them based on their roles in biological processes. Examples include Enzymatic Proteins, which catalyze biochemical reactions; Structural Proteins, contributing to cell and tissue structure; Contractile Proteins, involved in muscle contraction; Transport Proteins, facilitating substance movement across membranes; Regulatory Proteins, controlling cell functions; Respiratory Proteins, participating in respiration; Digestive Proteins, aiding in food digestion; Toxin Proteins, found in venoms and cell membranes; and Storage Proteins, storing specific elements like iron or copper due to multiple binding sites.

Properties of fatty acids that effect its chemical and and physical properties?

The properties of fatty acids, including chain length and degree of saturation, significantly impact their chemical and physical characteristics. Longer fatty acyl chains and fewer double bonds result in lower water solubility. Formation of esters, such as in glycerides, influences chemical behavior. Fatty acids form salts with alkali metals, producing soaps. Additionally, variations in chain length and saturation affect melting points and solubility in water. These properties, influenced by hydrocarbon chain structure and functional groups, contribute to the diverse roles of fatty acids in biological and chemical processes.

Explain Hydrogenation of fats.

Hydrogenation of fats involves converting carbon-carbon double bonds to single bonds by adding hydrogen. This process increases the melting point, transforming oils into solid fats at room temperature. Common in Banaspati ghee production, inexpensive oils like cottonseed oil undergo hydrogenation, becoming edible solid fats. The reaction occurs under high hydrogen pressure, catalyzed by nickel or copper, and heat. This industrial process alters the physical properties of fats, providing desirable characteristics for various food applications.

Write formation of banaspati ghee.

Banaspati ghee is formed through the hydrogenation of inedible and inexpensive oils like cottonseed oil. This industrial process involves converting carbon-carbon double bonds to single bonds by adding hydrogen, increasing the melting point and transforming oils into solid fats. Catalyzed by finely divided nickel or copper and heat, the hydrogenation is done under high hydrogen pressure, resulting in the production of edible solid fats used in Banaspati ghee manufacturing.

Functional groups of pyrimidines and purines.

The most important functional groups of pyrimidines and purines are

1) ring nitrogens, 2) carbonyl groups, and 3) exocyclic amino groups.

Define enzymes kinetics.

Enzyme kinetics is the study of the rates at which enzymes catalyze chemical reactions. It involves analyzing the factors influencing reaction rates, understanding enzyme-substrate interactions, and determining kinetic parameters such as K_m (Michaelis constant) and V_{max} (maximum reaction velocity).

Write properties of serine and threonine.

Serine and threonine, classified as aliphatic, polar, uncharged amino acids, possess polar hydroxyl groups in their side chains. These hydroxyl groups serve as sites for attaching phosphate groups. Additionally, the hydroxyl group in serine and threonine acts as a site for oligosaccharide chain attachment in glycoproteins, facilitating various biological functions.

Action of acids on carbohydrates.

Acids hydrolyze polysaccharides into monosaccharides, but concentrated acids decompose monosaccharides. Pentoses, under acidic conditions, yield the cyclic aldehyde furfural, while hexoses produce hydroxymethyl furfural. Furfural compounds formed can condense with organic phenols, like α -naphthol, creating colored compounds. This principle is utilized in tests for sugar

detection, such as Molisch's and Selivanoff's tests, which rely on the characteristic colors formed through the condensation of furfurals with phenols in the presence of acids.

Define Induced fit:

The induced fit model in biochemistry describes how enzymes and substrates interact during a chemical reaction. It emphasizes that the enzyme's active site isn't a rigid structure but can change its shape slightly when a substrate binds. This dynamic adjustment optimizes the fit between the enzyme and substrate, enhancing the catalytic efficiency of the enzyme in facilitating the chemical reaction. It's like a molecular handshake that fine-tunes for better coordination, ensuring the most effective conversion of substrates into products.

write isomer classification:

Isomers are classified into structural isomers, which differ in the arrangement of atoms, and stereoisomers, which have the same structural formula but differ in spatial arrangement. Stereoisomers include geometric isomers, which differ in cis-trans arrangements, and optical isomers, which exhibit different optical activities.

Write glycerol properties:

Glycerol, a triol, is a colorless, odorless, and sweet-tasting liquid. It is hygroscopic, miscible with water, and has a high boiling point. Glycerol is a key component in the synthesis of lipids and serves as a humectant, drawing moisture, making it valuable in various applications like skincare and pharmaceuticals.

Function of t RNA:

Transfer RNA (tRNA) plays a crucial role in protein synthesis by carrying specific amino acids to the ribosome during translation. Each tRNA molecule has an anticodon that base-pairs with a complementary codon on mRNA, ensuring the accurate incorporation of amino acids into the growing polypeptide chain.

Calculate the pKa of lactic acid, given that when the concentration of lactic acid is 0.01M and the concentration of lactate is 0.087 M, the pH is 4.80

$$\text{pH} = \text{pKa} + \log \frac{[\text{lactate}]}{[\text{lactic acid}]}$$

$$\text{pKa} = \text{pH} - \log \frac{[\text{lactate}]}{[\text{lactic acid}]}$$

$$= 4.80 - \log 0.087/0.01$$

$$= 4.80 - \log 8.7$$

$$= 4.80 - 0.94$$

$$= 3.9$$

Rate of reaction with substrate concentration

The rate of an enzyme-catalyzed reaction shows a direct relationship with substrate concentration, increasing until it reaches maximal velocity (V_{max}). As substrate concentration rises, the reaction rate increases, indicating more substrate molecules binding to available enzyme sites. However, at high substrate concentrations, the rate plateaus, reaching saturation as all enzyme binding sites become occupied. This phenomenon signifies that the enzyme is working at its maximum capacity, and further increases in substrate concentration do not enhance the reaction rate.

Rancidity types detail

Rancidity refers to the development of undesirable odors and flavors in fats and oils due to the decomposition of lipids. There are two main types of rancidity:

1= Oxidative Rancidity: Oxidative rancidity occurs when fats undergo oxidation, resulting in the formation of aldehydes, ketones, and dicarboxylic acids that contribute to an objectionable taste

and odor. This type of rancidity requires the presence of oxygen in the air and can be prevented by adding antioxidants like vitamin E to foods.

2= Hydrolytic Rancidity: Hydrolytic rancidity, on the other hand, is caused by the slow hydrolysis of fats. In the case of fats like butter, this leads to the release of short-chain fatty acids that are volatile and have a rancid taste and odor. Unlike oxidative rancidity, hydrolytic rancidity does not require the presence of oxygen.

What is the difference between Fats and oil

The main difference between fats and oils lies in their physical state at room temperature. Fats are typically solid, while oils are liquid. This distinction arises from the fatty acid composition; fats contain a higher proportion of saturated fatty acids, making them solid, whereas oils have more unsaturated fatty acids, keeping them in a liquid state.

Activation energy rate of reaction:

Enzymes reduce the activation energy required for chemical reactions by stabilizing the transition state through the formation of enzyme-substrate complexes. This enables reactions to proceed more rapidly. Enzymes achieve this by bringing reactants closer, providing a favorable microenvironment, or participating directly in the reaction. Lowering activation energy is crucial for increasing reaction rates, highlighting the essential role of enzymes in biological processes.

Define melting temperature:

When DNA is heated, the temperature at which one half of the helical structure is lost is defined as the melting temperature.

Digestive protein and their example:

These proteins are digestive enzymes which digest our food materials such as carbohydrates, proteins, lipids and include Amylase Pepsin Lipases etc.

zero order reaction:

A zero-order reaction is a chemical reaction with a constant rate independent of reactant concentration. The rate equation, represented as "Rate = k," indicates that the reaction proceeds at a fixed rate, unaffected by the concentration of reactants.

Work of protease enzym and dehydrogenase:

Proteases hydrolyze proteins, and dehydrogenases remove hydrogen atoms.

Why mRNA is heterogeneous:

mRNA is heterogeneous due to variations in abundance, size (ranging from 500 to 6000 nucleotides), base sequence, and stability. This diversity arises from different mRNA molecules formed with DNA template strands during transcription, encompassing untranslated regions at the 5' and 3' ends, along with the presence of a 5' cap and a poly A tail at the 3' end.

Describe the km menten kinetics equation:

$K_m = [S]$, when $V_0 = 1/2 V_{max}$.

This is a very useful, practical definition of K_m : K_m is equivalent to the substrate concentration at which V_0 is one-half V_{max} . The equation describes the kinetic behavior of a great many enzymes, and all enzymes that exhibit a hyperbolic dependence of V_0 on $[S]$ are said to follow Michaelis- Menten kinetics.

Describe pyrimidine bases:

Pyrimidine bases are nitrogenous compounds found in nucleic acids, specifically DNA and RNA. They include cytosine, thymine (in DNA), and uracil (in RNA). Pyrimidines have a single-ring structure and participate in base pairing to form the genetic code, influencing genetic information transmission.

Sources of lipids

Lipids are derived from various sources, including animal fats (meat, dairy), plant oils, nuts, seeds, and seafood. Additionally, lipids can be synthesized within the body as a part of metabolic processes.

What is titration curve of glycine? (long)

The titration curve of glycine reveals key information: pKa values of its ionizing groups (2.34 for -COOH, 9.60 for -NH₃⁺), indicating acidic and basic strengths. Two buffering regions emerge: one near pH 2.6 and another around pH 9.60. Glycine is an effective buffer within these zones. Notably, at physiological pH (around 7.4), glycine lacks buffering capacity. The curve also illustrates glycine's net charge at specific pH values, highlighting its dipolar form at pH 5.97, fully ionized but electrically neutral.

Component of nucleotides:

Nucleotides, the building blocks of nucleic acids, consist of three components: a phosphate group, a pentose sugar (ribose in RNA or deoxyribose in DNA), and a nitrogenous base (adenine, thymine, cytosine, guanine, or uracil). These components combine to form the structure of nucleotides in DNA and RNA.

Result of reduction of glutamate:

Removal of glutamate residue from LTC₄ (4 is subscript) forms LTD₄, LTD₄ on removal of glycine becomes LTE₄:

What is Svedberg unit:

The Svedberg unit is a measure of sedimentation velocity used to define ribosomal subunits. It correlates with the molecular weight and shape of compounds. In the context of ribosomal RNA, bases include adenine, guanine, cytosine, uracil, and pseudouridine.

Cofactor and coenzyme:

Cofactors are inorganic substances or metal ions essential for enzyme activity. Coenzymes are organic, non-protein molecules that aid enzyme function, often transferring chemical groups. Both cofactors and coenzymes assist enzymes in catalyzing biochemical reactions within cells.

Equilibrium constant for reversible reaction:

The equilibrium constant (K) for a reversible reaction quantifies the ratio of product concentrations to reactant concentrations at equilibrium. It reflects the tendency of a reaction to reach a stable state and is expressed as [products]/[reactants] at equilibrium.

nucleotides as energy currency of cell:

Energy currency: Nucleotides play an important role as "energy currency" in the cell. • Nucleoside tri and diphosphates such as ATP and ADP are the principal donors and acceptors of phosphoryl group in metabolism. • By doing this, they play a key role in the energy transduction. • This energy is used in almost every energy requiring process in the body, such as; • Muscle contraction, • Transmission of nerve impulse, • Transports of nutrients across cell membrane • Motility of spermatozoa • And many more energy dependent processes.

Why glycine does not occur in alpha helix:

Glycine lacks a side chain and its small size allows excessive flexibility, disrupting the regular α -helix structure. In an α -helix, steric hindrance between adjacent glycine residues would destabilize the helical conformation, making it energetically unfavorable.

Difference between Template and Non template:

The template strand is the DNA sequence copied during mRNA synthesis, guiding mRNA formation. The non-template (coding) strand has a base sequence directly mirroring the mRNA, representing codons translated into proteins. The template is 3'-5', while the non-template is 5'-3'.

Three main function group of purine and pyrimidine:

Purines and pyrimidines serve as the fundamental building blocks of nucleic acids. Their three main functional groups include providing the structural framework for DNA and RNA, participating in energy transfer (ATP, GTP), and serving as precursors for various coenzymes and signaling molecules.

Polysaccharide and example:

A polysaccharide is a complex carbohydrate composed of multiple monosaccharide units linked together by glycosidic bonds. An example is starch, found in plants, serving as a storage form of glucose. It provides a sustained energy source upon digestion.

Difference small km and large km:

In enzymology, K_m (Michaelis constant) measures substrate concentration at half-maximal reaction velocity. A small K_m indicates high enzyme affinity for the substrate, requiring low substrate concentration for efficient catalysis. A large K_m implies lower affinity, demanding higher substrate concentrations for comparable reaction rates.

No. Of H⁺ Ions in a 0.1M solution of NaOH

In a 0.1M solution of NaOH, which is a strong base, there are 0.1 moles of hydroxide ions (OH⁻) per liter. Since NaOH fully dissociates, the number of H⁺ ions is equal to the number of OH⁻ ions, making it 0.1 moles per liter.

Monoenoic Acids:

Monoenoic acids are fatty acids with a single double bond in their hydrocarbon chain. Examples include oleic acid.

Dienoic Acids:

Dienoic acids have two double bonds in their hydrocarbon chain. Linoleic acid is an example, crucial for human health.

Isomerization:

Isomerization is the process of rearranging molecular structure, altering the position of double bonds. It influences the properties and functions of fatty acids.

Abbreviation of Methionine, Arginine, Alanine, asparagine, Glycine?

1= Methionine: Met=M

2= Arginine: Arg=R

3= Alanine: Ala=A

4= Asparagine: Asn=N

5=Glycine=Gly=G

Structure of glycine:

Glycine is the simplest amino acid with a hydrogen atom as its side chain. It consists of an amino group, a carboxyl group, and a hydrogen atom as its side chain. Its structure contributes to the diverse composition of proteins.

Role of proteins in respiration with example.

Proteins play crucial roles in respiration. Enzymes like cytochrome c oxidase facilitate electron transport chain reactions. Myoglobin and hemoglobin transport oxygen, while carbonic anhydrase helps in carbon dioxide transport and pH regulation.

Simple representation of Penta deoxy ribonucleic acid.

A simple representation of Penta deoxyribonucleic acid would involve a chain of five deoxyribonucleotide units, each containing a phosphate group, deoxyribose sugar, and one of the four nitrogenous bases (adenine, thymine, cytosine, or guanine), forming a linear structure.

Define reaction velocity and example.

The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per unit time; • Velocity is usually expressed as μmol of product formed per minute. The rate of enzyme catalyzed reaction increases as the substrate concentration increases until it reaches a maximum velocity (V_{max}). This maximum velocity means that all the available binding sites on the enzyme are occupied by substrate molecules, and further increases in substrate concentration do not affect the reaction rate.

Describe perks of quaternary structure.

The perks of quaternary protein structure lie in increased stability and functionality. Larger size enhances interactions among amino acid residues, making unfolding and refolding challenging. Multisubunit structures offer advantages like cooperativity in ligand binding, high-affinity binding sites for large molecules, and diverse subunit activities collaborating for specific functions, exemplified by enzyme complexes like Pyruvate Dehydrogenase

Subclass of histone proteins:

Histone proteins are categorized into two main subclasses: the core histones (H2A, H2B, H3, H4), forming the histone octamer around which DNA wraps, and the linker histones (H1), involved in stabilizing higher-order chromatin structures and regulating gene expression.

One letter symbols for amino acids:

One-letter symbols represent amino acids in biochemistry. Examples include:

A: Alanine, C: Cysteine, D: Aspartic acid, E: Glutamic acid, F: Phenylalanine, G: Glycine, H: Histidine, I: Isoleucine, K: Lysine, L: Leucine, M: Methionine, N: Asparagine, P: Proline, Q: Glutamine, R: Arginine, S: Serine, T: Threonine, V: Valine, W: Tryptophan, Y: Tyrosine

What is the function of relatively low amount of lipids in our body?

While lipids provide energy storage, their relatively low amount in the body contributes to structural roles in cell membranes, insulation, and hormone synthesis. Maintaining a balance is crucial for proper cell function, insulation against temperature changes, and signaling pathways essential for overall health.

Write two examples of hydrolases?

Two examples of hydrolases are lipases (catalyze lipid hydrolysis) and amylases (facilitate the hydrolysis of starch into sugars).

What is amyloidosis?

Amyloidosis is a disorder characterized by the abnormal accumulation and deposition of insoluble protein aggregates called amyloids in tissues and organs, impairing their structure and function. It can be systemic or localized.

Keratins:

Keratins are fibrous structural proteins constituting hair, nails, and epidermal cells. They provide strength, rigidity, and protective functions to these tissues.

Chargaff's Rules:

Chargaff's rules state that in DNA, the amount of adenine (A) equals thymine (T), and guanine (G) equals cytosine (C). This underpins the complementary base pairing in DNA.

Polyunsaturated Fatty Acids Examples:

Polyunsaturated fatty acids include omega-3 (e.g., alpha-linolenic acid) and omega-6 (e.g., linoleic acid) types found in various oils and fish.

Metabolism Pathways of Arachidonic Acid:

Arachidonic acid undergoes metabolism via the cyclooxygenase, lipoxygenase, and cytochrome P450 pathways, producing diverse bioactive compounds, including prostaglandins and leukotrienes.

Saponification and End Products:

Saponification is the hydrolysis of ester bonds in triglycerides to produce glycerol and soap. The end products are fatty acid salts (soap) and glycerol.

Advantages of Lineweaver-Burk Plot:

The Lineweaver-Burk plot simplifies enzyme kinetics analysis. Its linear representation facilitates determining V_{max} and K_m values, aiding in understanding enzyme-substrate interactions and inhibitor effects with clarity.

Platelet-activating factor?

Platelet-activating factor (PAF) is an atypical ether glycerophospholipid. It contains a saturated alkyl group and an acetyl residue, inducing powerful blood clotting and inflammatory responses. PAF, produced by various cells, activates platelets, stimulates immune cells, and plays a crucial role in clot formation and microbial defense.

Aromatic amino acids with examples?

Aromatic amino acids contain a benzene ring. Examples include phenylalanine, tryptophan, and tyrosine. Phenylalanine contributes to protein synthesis and neurotransmitter production, tryptophan is a precursor for serotonin, and tyrosine is involved in catecholamine synthesis and thyroid hormone production.

How isozymes of hexokinase illustrates the significance of K_m ?

Hexokinase isozymes exhibit different K_m values for glucose. Hexokinase I has a low K_m , indicating high affinity for glucose, suitable for glucose phosphorylation in tissues with low glucose levels. Hexokinase IV (glucokinase) has a higher K_m , suited for liver's role in glucose homeostasis.